Homework Assignment 4

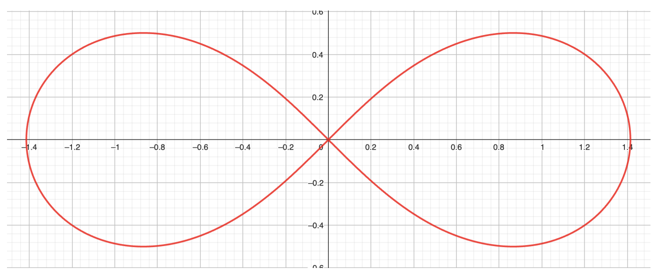
Show all your computations and justify your answers.

1. **(20 points)** Show that the curve $f(x) = 2\ln x + x^2$ has no tangent line parallel to $y = 2x$.

2. **(20 points)** Consider the function $f(x) = e^{2x}(\cos 3x - \sin 3x)$. Prove that $f''(x) - 4f'(x) + 13f(x) = 0$ for any x in $\mathbb{R}$.

3. **(20 points)** Let f be a differentiable function on $\mathbb{R}$ such that $f\left(\frac{\pi}{2}\right) = -2\pi$, $f'\left(\frac{\pi}{2}\right) = 2$ and $f'\left(\pi(2-\pi)\right) = -1$. For $g(x) = f((x\sin x - 1)f(x))$, find $g'\left(\frac{\pi}{2}\right)$.

4. **(20 points)** Find all the horizontal tangent lines of the curve $(x^2 + y^2)^2 = 2x^2 - 2y^2$.



5. **(20 points)** Find the derivative of $f(x) = \frac{\log_2(x^2+2)}{2\sqrt{x}}$.